Parkinson's Freezing of Gait Prediction

EVENT DETECTION FROM WEARABLE SENSOR DATA

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Project

- Parkinson’s disease – progressive neurodegenerative disorder, affecting motor system and gait
- Affects ~300 000 people in Germany
- Symptoms: tremor, rigidity, slowness, balance and gait problems

Freezing of gait:

- Sudden, brief inability to move feet forward
- Patient feels "glued to the floor“
- Lasts seconds but causes falls, injuries, loss of independence

Three Types: Start Hesitation, Turn, Walking

Project

- Acceleration data from a lower back Inertial Measurement Unit
- 136 patients, age range: 28 - 94 years
- 2 datasets – collected in the lab and at home
  - Different patients in both datasets
  - 833 files from 62 patients in the lab dataset
  - 137 files from 45 patients in the home dataset
- 3 types of FoG (StartHesitation, Turn, Walking):
  - High class imbalance (Turn FoG > ¾ of all FoG events)
- Goal:
  - To detect FOG episodes for the tdcsfog and defog series
  - To detect the type of FOG episodes (StartHesitation, Turn , Walking)

Baseline model

Logistic regression

- Validation AP:
  - StartHesitation: 0.1346
  - Turn: 0.4116
  - Walking: 0.0664

Place 1212 / 1379 kaggle

Private Score	Public Score
0.105048	0.176136

Feature engineering #1

Raw Data (3):

- AccV, AccML, AccAP (direct sensor readings)

Derived (4):

- Magnitude: $\text{sqrt}(\text{AccV}^2 + \text{AccML}^2 + \text{AccAP}^2)$
- 3 jerk features: rate of change

Total: 7 features, minimal statistical context

LSTM #1

Model: Bidirectional LSTM with Attention

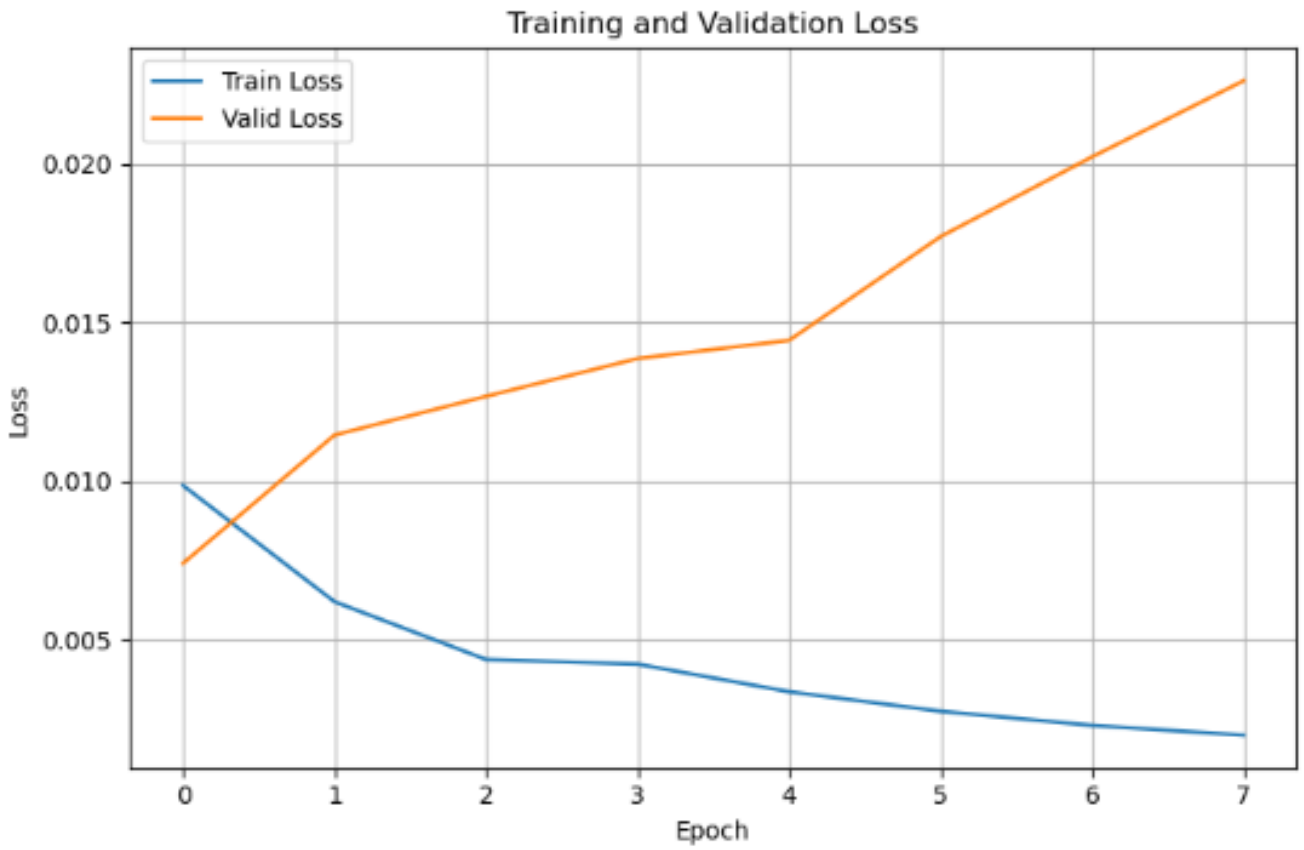
- Input: (batch_size = 512, 96, 7):
  - 7 features: 3 raw + 4 derived
  - Sliding window with 72-step past, 24-step future
- Architecture: BiLSTM (256 hidden units, 2 layers)
- Attention Mechanism
- Dense (256, ReLU)
- Dropout (0.4)
- Output layer (3) → [StartHesitation, Turn, Walking]

Results (LSTM #1)

- Validation Score (AP): **0.2331**
- Per class AP:
  - StartHesitation: 0.031
  - Turn: 0.646
  - Walking: 0.022

kaggle Place 89 / 1379 (bronze)

Private Score	Public Score
0.294175	**0.254824**



Feature engineering #2

Whole timeseries compressed to 2048 points

Basic Statistics (5 features × 3 axes = 15):

- Mean: Average movement intensity over window
- Max: Peak acceleration (sudden movements)
- Min: Minimum acceleration (stillness periods)
- Median: Central tendency, robust to outliers
- Std: Movement variability

Extended Statistics (6 features × 3 axes = 18):

- p15, p30, p45, p60, p75, p90

Total: 33 features

LSTM #2

- Model: Bidirectional LSTM
- Input: (batch, 2048, 33):
  - 2048 timesteps (20 seconds @ 100Hz)
  - 33 features: 15 basic statistics + 18 percentile features
- Architecture: BiLSTM (1024 units, 4 layers)
- Dense (128, SELU)
- Output layer (4, softmax) → [StartHesitation, Turn, Walking, NoEvent]

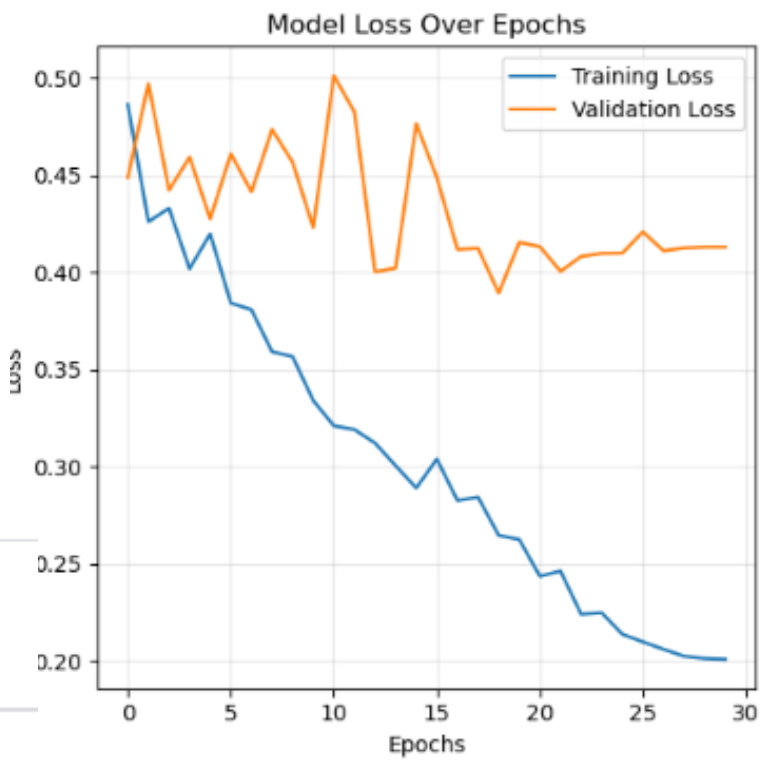
Results (LSTM #2)

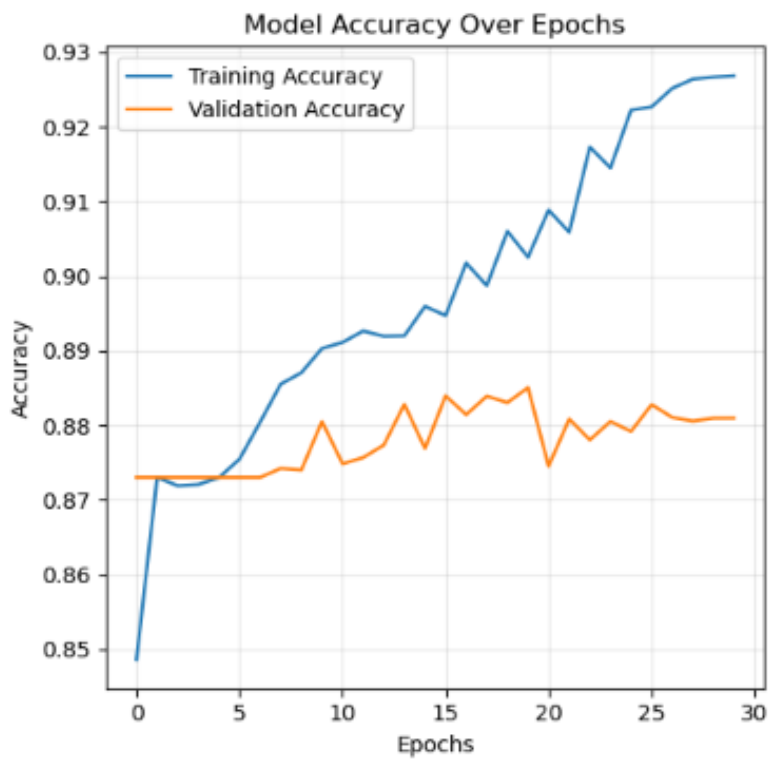
- Event Detection AP (any FOG): **0.4456**

- Per class AP:
  - StartHesitation: 0.0338
  - Turn: 0.4432
  - Walking: 0.0264

kaggle

Private Score	Public Score
0.343093	**0.339865**





Key improvements

- Better feature engineering (7 raw data features vs 33 data-derived features)
- Temporal context (1 sec window vs whole recording compressed to 20 sec)
- Model depth (2 layers vs 4 layers)

Kaggle ranking

#	△	Team	Members	Score
1	▲ 3	Baurzhan Urazalinov		0.514025
2	▼ 1	waiwai		0.450906
3	▲ 2	Stochoshi		0.436426
10	▲ 12	James Day		0.347748
11		karsae		**0.343093**
12	▲ 50	Ismail		0.341435

What would I do next?

- Convert IMU acceleration data to images (Signal-to-Image encoding)
- Use machine learning models (e.g., CNNs) for image classification

Journals & Magazines > IEEE Access > Volume: 12

Signal to Image Conversion and Convolutional Neural Networks for Physiological Signal Processing: A Review

Publisher: IEEE Cite This PDF

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